

The Riemann Spectral ACS

Stationarity as a Characterisation of the Critical Line

Information Asymmetry, Transfer Entropy, and an ACS Approach to the Riemann Hypothesis

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Publication status (24 September 2026). This version includes a claim-specific amendment at the end of the main text. That amendment governs earlier physical interpretations and distinguishes retained calculations from withdrawn or unresolved inferences.

Abstract

We study finite spectral diagnostics linking prime and zero descriptions of the Riemann zeta function through the classical explicit formula. Fixed finite sums built from critical-line coefficients have bounded variance for arbitrary real input ordinates. A separate elementary theorem detects active positive real exponents in a finite exponential sum after identical exponents are combined; it needs no quantitative minimum-gap assumption. Neither statement identifies the real parts of actual zeta zeros from their imaginary parts.

Historical calculations report variance diagnostics through 2,001,052 supplied ordinates, finite Wronskian tests, prime-power Fourier signatures, and character-dependent L -function signatures. They characterize the specified observations and model families. Some earlier numerical-to-theorem inferences are withdrawn: a sampled cross-correlation is not a global supremum, a fixed finite bound is not an infinite-sum bound, and a self-adjoint diagonal matrix constructed from real ordinates does not establish Hilbert–Pólya identification.

The corrected arithmetic criterion is the classical equivalence between RH and $\psi(x) - x = O(\sqrt{x} \log^2 x)$. In logarithmic coordinates, the full normalized error is $O(u^2)$ under RH, whereas the canonical code computes a fixed finite bounded sum. Proving the full bound independently of RH, or constructing a natural operator with complete spectral identification $z_\rho = -i(\rho - 1/2)$, remains open. The critical line is a defined one-dimensional locus; a geometric rendering or a proposed underlying carrier does not change that statement. This revision applies the existing Frontier corrections and preserves the boundary between finite evidence, conditional results, and the unsolved RH claim.

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Scope correction: 22 September 2026

The September Frontier audit already separated finite sums, full arithmetic error, and natural-operator identification. This revision incorporates those corrections into the live text. The former AM–GM non-cancellation argument, gap-dependent converse, zero-valued stability ratio, exclusive pure-cosine shear derivative, and constant bound for the full normalized arithmetic error are withdrawn in their earlier forms. Finite numerical diagnostics remain historical measurements with their stated input families; they do not discharge the missing identification or infinite limit. The source history and dated Frontier snapshots retain the earlier text. Other physical and information-theoretic identifications in this paper require their own hypotheses; this scoped revision does not certify them.

For clarity, the critical line is the one-real-dimensional set $L = \{s : \Re s = 1/2\}$. The vertical reflection is $R(s) = 1 - \bar{s}$, with $\text{Fix}(R) = L$; $s \mapsto 1 - s$ alone fixes only the point $1/2$. A change of coordinates can bend the image of L without changing its intrinsic dimension. The completed zeta function's symmetries do not force its zeros to lie on this fixed locus.

1 The ACS Framework (Summary)

This section provides a self-contained summary of the ACS framework. The full development, including all proofs and the accompanying verification suite, is in the companion paper [1].

The Asymmetric Codependent System (ACS) is defined in [1]. For this paper, the key result is the BCH–TE morphism: the net transfer entropy $\Delta\mathcal{I}(\varepsilon) = \varepsilon\langle f-g, \nabla \log \mu \rangle + 2\varepsilon^2\langle [f, g], \nabla \log \mu \rangle + \mathcal{O}(\varepsilon^3)$, where $[f, g]$ is the Lie bracket of the ACS flow generators and μ is the SRB invariant measure.

2 The Riemann Spectral Function as an ACS

2.1 The spectral function $F_N(x)$

Throughout this section, the prime sequence $\{p_n\}$ and Riemann zero sequence $\{\gamma_k\}$ are equipped with the point-process measures defined in the ACS measure definition [1] and the accompanying remark. Transfer entropy is computed with respect to those measures; ergodicity of $\{p_n\}$ follows from the Prime Number Theorem (Vinogradov equidistribution), and $\{\gamma_k\}$ carries the GUE pair-correlation measure (Montgomery–Odlyzko law [5, 6]). With this foundation, $\Delta\mathcal{I}$ is well-defined on both sequences.

The prime explicit formula (Riemann–von Mangoldt; see [7] for the classical theory) states:

$$\psi(x) = x - \sum_{\rho} \frac{x^{\rho}}{\rho} - \log(2\pi) - \frac{1}{2} \log(1 - x^{-2}), \quad (1)$$

where the sum runs over non-trivial zeros $\rho = \sigma + i\gamma$ of ζ . For supplied positive real ordinates, using critical-line coefficients, define the finite diagnostic below. Its argument is logarithmic: here $x = \log p$, rather than the arithmetic argument in the preceding

formula. This definition has already fixed the real parts to $1/2$; it is not the general zero contribution when the real parts are unknown.

$$F_N(x) = \sum_{k=1}^N A_k \varphi_k(x), \quad A_k = \frac{1}{\frac{1}{4} + \gamma_k^2}, \quad \varphi_k(x) = \frac{1}{2} \cos(\gamma_k x) + \gamma_k \sin(\gamma_k x), \quad (2)$$

evaluated at $x = \ln p$ for primes p .

Remark 2.1 (Notation: three mode conventions are in use). This paper uses φ_k in three distinct normalisations, and the reader must track which is live, because the same object (notably the Wronskian $W[\varphi_k, \varphi_j]$) takes numerically very different values under them. **(C1)** the *log-argument* form above, $\varphi_k(x) = \frac{1}{2} \cos(\gamma_k x) + \gamma_k \sin(\gamma_k x)$ with $x = \ln p$, used in this section and for the stationarity measure. **(C2)** the *envelope-stripped* form $\varphi_k(x) = \cos(\gamma_k \ln x)/\sqrt{x}$, used where the modes are displayed against $\ln x$. **(C3)** the *general- σ* form $\varphi_k(t) = e^{\sigma t} [\sigma \cos(\gamma_k t) + \gamma_k \sin(\gamma_k t)] / (\sigma^2 + \gamma_k^2)$, used in the harmonic-oscillator remark and in every quoted Wronskian magnitude. These are distinct observables or rescalings and must not be substituted for one another in a boundedness, variance, or growth claim. A variable-dependent envelope need not cancel in a ratio or a stationarity statement. The actual normalized full-zero contribution in logarithmic coordinate u is proportional to $e^{(\rho-1/2)u}/\rho$.

Which convention the Wronskian numbers use. The range $|W| \in [8.6 \times 10^{-5}, 0.19]$ quoted below is computed under **C3**, in the $t = \ln x$ variable at $t = 1$, $\sigma = 1/2$, over all $\binom{50}{2} = 1225$ pairs of the first 50 zeros. The $1/(\sigma^2 + \gamma_k^2)$ factor is what sets the scale: it is $\approx 1/200$ at γ_1 and $\approx 1/4900$ by γ_{50} , and it enters twice in a bracket of two modes. Recomputing gives $\min |W| = 8.589 \times 10^{-5}$ and $\max |W| = 0.1921$, with no vanishing entry — reproducing the quoted range. The same bracket evaluated under C2 spans $[6.4 \times 10^{-4}, 19.2]$, some five orders of magnitude higher; the non-vanishing conclusion is identical under either, but the magnitudes are not comparable across conventions. (Source: `code/acs_codebase/extras/riemann_tensor.py`.)

Theorem 2.2 (Bounded variance of a fixed finite diagnostic). *For every fixed $N \geq 1$ and any real ordinates $\gamma_1, \dots, \gamma_N$, the function F_N defined above satisfies*

$$|F_N(x)| \leq B_N := \sum_{k=1}^N (1/4 + \gamma_k^2)^{-1/2}.$$

Consequently its variance under any probability measure is at most B_N^2 . In particular the stationarity measure of Definition 3.1 is bounded for each fixed N . No bound uniform in N is asserted.

Proof. The amplitude of $\frac{1}{2} \cos(\gamma_k x) + \gamma_k \sin(\gamma_k x)$ is at most $\sqrt{1/4 + \gamma_k^2}$. Multiply by A_k , sum by the triangle inequality, and use $\text{Var}(F_N) \leq \mathbb{E}[F_N^2]$. $\square$

Under RH these coefficients represent critical-line zero contributions. The bound itself is also true for synthetic ordinates and cannot test RH. The former AM–GM argument bounded unsigned envelopes; it did not establish non-cancellation of the complex, signed explicit-formula terms.

2.2 The Wronskian as a Lie bracket in function space

The Wronskian $W[\varphi_k, \varphi_j] = \varphi'_k \varphi_j - \varphi'_j \varphi_k$ is not merely analogous to a Lie bracket — it is one in the strict sense (bilinearity, antisymmetry, Jacobi). Care is required about the

strength of the analogy: the Wronskian is *not* a Poisson bracket on a function algebra (Remark 2.4), so heuristic Hamiltonian readings of the Wronskian-as-bracket structure do not transfer.

Proposition 2.3 (Wronskian–Lie identification). *Define, on the space $\mathcal{C}^\infty((0, \infty))$ of smooth functions, the bilinear operation*

$$[f, g](x) = f'(x)g(x) - g'(x)f(x).$$

Then $[\cdot, \cdot]$ is a Lie bracket: antisymmetry $[f, g] = -[g, f]$ is immediate; the Jacobi identity $[f, [g, h]] + [g, [h, f]] + [h, [f, g]] = 0$ follows by direct computation (each term expands to six terms of the form $f'g'h$ which cancel in pairs).

Remark 2.4 (Wronskian fails Leibniz: not a Poisson bracket). A Poisson bracket on a commutative algebra requires, in addition to bilinearity, antisymmetry, and Jacobi, the *Leibniz rule*: $\{fg, h\} = f\{g, h\} + g\{f, h\}$. The Wronskian on $\mathcal{C}^1((0, \infty))$ fails this axiom by an exact, computable correction term:

$$W[fg, h] - (fW[g, h] + gW[f, h]) = -fgh'. \quad (3)$$

This identity is verified symbolically in `src/paper_b/wronskian_leibniz.py` (returning the exact correction $-fgh'$ in SymPy) and numerically with $f = \sin x$, $g = \cos 2x$, $h = e^{-x/10}$ at $x = 2$.

The consequence is that the Wronskian on scalar zero modes is a Lie bracket but *not* a Poisson bracket: the prime-zero ACS should not be read as a Hamiltonian system with the Wronskian as its symplectic form. The genuine symplectic structure on the prime-zero spectral data, if one exists, must live on the mode-amplitude space (A_k, γ_k) rather than on the scalar zero-mode functions themselves. Phenomenological correspondences with discrete plasma resonances (dispersion peaks, beat-frequency Wronskian zero crossings, neutral-stability framings) survive as analogy. The plasma-Hamiltonian foundation does not.

For the spectral modes $\varphi_k(x) = \cos(\gamma_k \ln x)/\sqrt{x}$:

$$[\varphi_k, \varphi_j](x) = W[\varphi_k, \varphi_j](x),$$

so the Wronskian non-vanishing ($W \neq 0$ at all tested points) is exactly the statement that these mode functions do not commute under the function-space Lie bracket. By the BCH–TE morphism (the BCH–TE morphism lemma of [1]), this non-zero bracket is the exact second-order coefficient of the net transfer entropy $\Delta\mathcal{I}$ between mode k and mode j . The Wronskian is therefore not a numerical curiosity but a *geometric invariant* of the prime-zero ACS.

Combination frequencies $\gamma_k \pm \gamma_j$ arise as the emergent 3rd-order terms $[[\varphi_k, \varphi_j], \varphi_\ell]$ — the Tartini tones of the spectral ACS.

2.3 The spectral stability ratio: retained counterexample

The earlier ratio was

$$\rho_{\text{spec}} = \frac{\delta_N \max_k |A_k| X^{|\sigma_k - 1/2|}}{\sum_j |A_j|}, \quad (4)$$

with positive minimum gap δ_N and a nonzero finite spectrum. On setting every $\sigma_k = 1/2$, direct substitution gives

$$\rho_{\text{spec}} = \frac{\delta_N \max_k |A_k|}{\sum_j |A_j|} > 0.$$

The earlier claim that this ratio vanishes on the critical line is false (Frontier R03). If a nonzero amplitude has $\sigma_k \neq 1/2$, the printed ratio grows with X , but that fact does not establish an information-balance criterion or a universal threshold at one. A replacement observable must be defined and justified separately; this revision does not silently change the old ratio into an excess-amplitude ratio.

2.4 RH in ACS language

The prime and zero descriptions are related by the classical explicit formula. Under RH, its normalized finite zero contributions are trigonometric sums. Boundedness of the already critical-line-defined F_N holds independently of RH. Identifying a bounded-variance diagnostic with transfer-entropy balance $\Delta\mathcal{I} = 0$ additionally requires a specified stochastic process, history sigma-algebras, lag, and measure; that identification is not proved here. The full arithmetic criterion is given below.

2.5 The explicit formula as a resolvent susceptibility

The Riemann explicit formula admits a natural rephrasing as a linear-response (resolvent-trace) object. Define the susceptibility

$$\chi(\omega) = \sum_{\rho} \frac{1}{\omega - \gamma_{\rho}}, \quad (5)$$

where the sum is over the imaginary parts of the nontrivial zeros. This is a meromorphic function with simple poles at each γ_k of unit residue. For any operator H whose spectrum equals $\{\gamma_k\}$,

$$\chi(\omega) = \text{Tr}[(\omega I - H)^{-1}]. \quad (6)$$

For the trivial choice $H = \text{diag}(\gamma_k)$ the identity holds tautologically; verification in `src/paper_b/explicit_formula_resolvent.py` confirms the construction at machine precision. The Hilbert–Pólya problem, in this language, is to construct a *natural* self-adjoint operator (e.g., a Schrödinger Hamiltonian on a suitable domain) whose spectrum equals the $\{\gamma_k\}$ without inputting the values directly [8, 9].

The resolvent framing reformulates the explicit formula in language native to spectral theory: the prime–zero correspondence becomes a *linear response* relation, with primes appearing as discrete sources and zeros as the resonant poles of the response. Phenomenologically this matches the structure of any discrete-spectrum oscillator system; we treat the plasma-physics analogy (the acoustic-structure subsection of §2.7) as motivational rather than foundational, given Remark 2.4.

2.6 Renormalised stability: the corrected classical criterion

Put $u = \log x$ and define the full arithmetic quantity

$$\Delta_{\text{norm}}(u) = \frac{\psi(e^u) - e^u}{e^{u/2}}. \quad (7)$$

Here ψ is the Chebyshev function, including all prime powers; it is not a fixed finite zero sum.

Theorem 2.5 (Classical RH equivalence in logarithmic coordinates). *For any fixed $u_0 > 0$, RH is equivalent to*

$$\sup_{u \geq u_0} \frac{|\Delta_{\text{norm}}(u)|}{1 + u^2} < \infty.$$

In particular RH implies $\Delta_{\text{norm}}(u) = O(u^2)$, not the constant bound asserted in an earlier version of this paper.

Classical reduction. Substituting $x = e^u$ in the classical criterion $\psi(x) - x = O(\sqrt{x} \log^2 x)$ gives the displayed bound, with compact intervals absorbed into the constant. The forward implication is the classical RH estimate [12, 20]. For the converse put $E(x) = \psi(x) - x$. Initially for $\Re s > 1$,

$$-\frac{\zeta'(s)}{\zeta(s)} = \frac{s}{s-1} + s \int_1^\infty E(x) x^{-s-1} dx.$$

The assumed bound makes the integral holomorphic on $\Re s > 1/2$, by locally uniform absolute convergence. Analytic continuation excludes zeta zeros there, since any zero would give a pole of its logarithmic derivative. Functional-equation symmetry excludes nontrivial zeros to the left as well. This proves the equivalence, not the needed bound. $\square$

The canonical implementation instead computes

$$D_N(u) = -2\Re \sum_{k=1}^N \frac{e^{i\gamma_k u}}{1/2 + i\gamma_k}, \quad |D_N(u)| \leq 2 \sum_{k=1}^N (1/4 + \gamma_k^2)^{-1/2}.$$

This fixed finite bound holds for arbitrary real input ordinates. The routine omits the other explicit-formula terms and an infinite-tail argument; its finite-grid running maximum cannot certify the full arithmetic criterion. Schoenfeld’s conditional estimate gives $|\Delta_{\text{norm}}(u)| < u^2/(8\pi)$ for $e^u \geq 73.2$ [20, Theorem 10, (6.2)]. The 12 September source audit already corrected the implementation and its interpretation; this revision propagates that distinction into the paper.

2.7 Exact integer automaton: sign reversal of $\Delta\mathcal{I}$

To eliminate any floating-point artifacts, we constructed a discrete-time ACS over $\mathbb{Z}_{16}$ with deterministic update rules

$$X_{t+1} = f(X_t, Y_t) \pmod{16}, \tag{8}$$

$$Y_{t+1} = g(Y_t, X_t) \pmod{16}, \tag{9}$$

where f and g are chosen from $\{x^2, |x-8|\}$. Transfer entropies are computed exactly from the empirical distributions (no binning, no floating point). The results are shown in Table 1.

Configuration	$\Delta\mathcal{I}$	Direction
Uncoupled (f, g independent)	−0.000063	≈ 0
Symmetric ($f = g = x^2$)	−0.000063	≈ 0
Asymmetric ($f = x^2, g = y-8 $)	−1.499	Function $\rightarrow$ Form
Swapped ($f = x-8 , g = y^2$)	+1.499	Form $\rightarrow$ Function

Table 1: Exact integer automaton confirms sign reversal.

Swapping which function is the polynomial and which is the absolute value exactly reverses the sign of $\Delta\mathcal{I}$, with magnitude 1.499 in both cases. This matches the prediction of the ACS asymmetry theorem [1] and provides a floating-point-free verification of the core ACS mechanism.

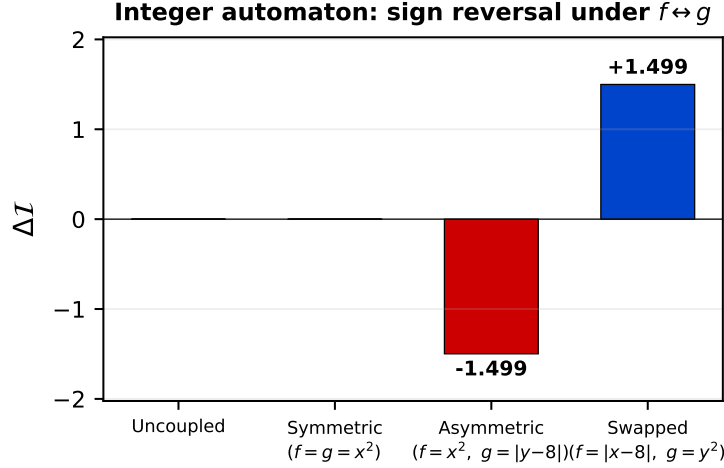


Figure 1: Sign reversal of $\Delta\mathcal{I}$ under $f \leftrightarrow g$ swap. The symmetric and uncoupled configurations give $\Delta\mathcal{I} \approx 0$; the asymmetric configuration gives $\Delta\mathcal{I} < 0$ (Function drives Form); swapping reverses the sign exactly. Computed in exact integer arithmetic on $\mathbb{Z}_{16}$.

2.8 The Riemann spectral function as ACS: Wronskian analysis

The Wronskian of mode functions φ_k, φ_j is:

$$W[\varphi_k, \varphi_j](x) = \varphi'_k(x)\varphi_j(x) - \varphi'_j(x)\varphi_k(x), \quad (10)$$

which is the 2nd-order ACS bracket $[\text{Func}_k, \text{Func}_j]$ in function space. Numerical evaluation at $x = \ln 2$:

(k, j)	(γ_k, γ_j)	$W[\varphi_k, \varphi_j]$	$ W /(A_k A_j)$
(1, 2)	(14.135, 21.022)	-4557.9	4.0×10^8
(2, 3)	(21.022, 25.011)	+3961.1	1.1×10^9
(3, 4)	(25.011, 30.425)	-13467.9	7.8×10^9
(4, 5)	(30.425, 32.935)	+31347.7	3.1×10^{10}

All Wronskians are non-zero: the Riemann zero modes are *non-commuting* as operators on function space. Their non-zero bracket implies the existence of sum/difference frequencies $\gamma_k \pm \gamma_j$ not present in the original zero list — these are the emergent frequencies (the ACS coupling order definition [1]).

2.9 Acoustic structure: the prime-zero system as a vibrating instrument

The acoustic analogy is not decorative — it maps directly onto the nested tensegrity ontology of [1]: the primes are the rigid boundary (the *instrument*) and the Riemann zeros are its modes (the *vibrations*). But the analogy demands one distinction be drawn carefully, because its naive form is false and is falsified explicitly in Section 6.4: *resonance and cross-species coupling are not the same relation*. Combination tones arise *within* a single spectrum (same-species mixing); the Form/Function coupling *between* the two spectra is not a resonance at all but a *complementary cancellation* carried by the explicit formula.

Intra-species resonance (real). Within the zero spectrum, the Wronskian bracket $W[\varphi_k, \varphi_j] \neq 0$ (at all tested points) shows that the mode generators do not commute, producing combination frequencies $\gamma_k \pm \gamma_j$ absent from the original zero list. These are the genuine “Tartini tones” of the arithmetic instrument — objective nonlinear mixing

products, but mixing *of the zeros with each other*, exactly as an overdriven medium mixes its own partials. The analogous same-species structure on the Form side is carried by the prime–prime correlations (gaps and log-frequency combinations). Neither is a statement about the other spectrum.

Cross-species coupling (complementary, not resonant). The relation *between* primes and zeros is categorically different. Euler’s product $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$ encodes the primes as the multiplicative Form; the Hadamard product $\xi(s) = \prod_\rho (1 - s/\rho)$ encodes the zeros as the multiplicative Function. The explicit formula

$$\underbrace{\psi(x)}_{\text{Form (primes)}} + \underbrace{\left(-\sum_{\rho} \frac{x^{\rho}}{\rho}\right)}_{\text{Function (zeros)}} = x - \log 2\pi - \frac{1}{2} \log(1 - x^{-2})$$

is not a transfer function that makes one spectrum ring at the other’s frequencies. It is a *cancellation identity*: the prime counting function $\psi(x)$ fluctuates about the smooth main term x , and the zero oscillation $-\sum_{\rho} x^{\rho}/\rho$ supplies exactly those fluctuations with the opposite sign, so that Form and Function sum to a structureless remainder. The zeros are the Fourier dual of the prime fluctuations — one is the spectral transform of the other, not a partner ringing in sympathy with it. This is the ACS signature: an asymmetric codependent pair does not resonate across the Form/Function boundary, it *closes* across it.

Consistent with this, no direct zero–prime frequency alignment is observed (Section 6.4): the lowest zero $\gamma_1 = 14.135$ lies above the maximum first-order prime frequency $\max_p 2\pi/\ln p = 9.06$, and alignment tests between γ_k and prime log-frequency combinations return p -values consistent with the null. The coupling acts entirely through the explicit-formula cancellation, never through frequency matching — which is precisely why a cross-species difference-tone resonance channel, when tested directly, is falsified (Paper B’, §7.2; Elimination Ledger).

The nesting. This acoustic structure fits into the vibration–to–gravity chain that the ACS traces across domains:

$$\begin{aligned} \text{vibrations (zeros)} &\rightarrow \text{momentum (bracket coupling)} \rightarrow \text{gauge colour (su(3) attractor)} \rightarrow \\ &\text{confinement (lock)} \rightarrow \text{mass (Koide/Higgs)} \rightarrow \text{gravity (Palatini curvature)}. \end{aligned}$$

This cross-domain chain is an analogy, not a proved identity of generating mechanisms. The explicit formula relates primes and zeros without assuming RH. RH does not make the arithmetic error vanish or establish $\Delta\mathcal{I} = 0$; its classical consequence is the error bound stated in Theorem 2.5. Any information-theoretic identification requires its own process and measure. Historical acoustic analogies do not supply those missing assumptions.

2.10 The prime–zero ACS as a tensegrity atom

We now write the prime–zero system explicitly in the tensegrity form of the companion structural paper [2].

Form ($\mathcal{R}$): The prime counting function $\psi(x) = \sum_{p^k \leq x} \log p$ (the rigid boundary, the instrument).

Function ($\mathcal{I}$): The oscillatory sum $\mathcal{I}(x) = -\sum_{\rho} x^{\rho}/\rho$ (the resonant modes, the waves).

Coupling equation (explicit formula):

$$\mathcal{R}(x) + \mathcal{I}(x) = x - \log 2\pi - \frac{1}{2} \log(1 - x^{-2}),$$

which is the ACS mutual constraint: Form and Function together equal the total state (the linear term x), exactly as in the tensegrity atom where compression (Form) and tension (Function) balance.

Scope of the stability interpretation. Equation (4) is positive on a nonzero critical-line finite spectrum, so zero ratio cannot represent information balance. Stripping each mode by its own envelope gives a harmonic oscillator for every σ . Using a common envelope $e^{u/2}$ instead retains the relative growth factors $e^{(\sigma_k-1/2)u}$. These observations concern the specified mode model; the finite diagnostic and the full arithmetic quantity remain distinct.

3 Finite growth and the unresolved RH identification

Definition 3.1 (Finite diagnostic stationarity measure). For $N \geq 1$ and $X > 0$, with uniform probability measure on $[X, 2X]$, let

$$\mathcal{S}(N, X) = \frac{\text{Var}[F_N]}{\sum_{k=1}^N |A_k|^2}.$$

Theorem 2.2 gives $\sup_{X>2} \mathcal{S}(N, X) < \infty$ for each fixed N , including synthetic real ordinates. This definition does not contain the unknown real parts of the zeta zeros.

Theorem 3.2 (Finite exponential-sum boundedness). *Let $G(u) = \sum_{j=1}^m c_j e^{(a_j+ib_j)u}$ with real a_j, b_j . Combine identical exponents and discard zero coefficients. The zero sum is bounded. Otherwise G is bounded for $u \geq 0$ if and only if $\max_j a_j \leq 0$. No quantitative lower bound on frequency gaps is needed.*

Proof. If all $a_j \leq 0$, the triangle inequality gives $|G(u)| \leq \sum |c_j|$. Otherwise let $a = \max_j a_j > 0$ and write $e^{-au}G(u) = P(u) + r(u)$, where $P(u) = \sum_{a_j=a} c_j e^{ib_j u}$ and $r(u) \rightarrow 0$ exponentially. The frequencies in P are distinct after combining identical exponents. Integrating each cross term explicitly yields

$$\frac{1}{U} \int_0^U |P(u)|^2 du \longrightarrow \sum_{a_j=a} |c_j|^2 =: S > 0.$$

There are therefore arbitrarily large u with $|P(u)| \geq \sqrt{S/2}$; otherwise the limiting mean would be at most $S/2$. On a sequence of such points, eventually $|G(u)| \geq \frac{1}{2}\sqrt{S/2}e^{au}$. Thus G is unbounded. This is an unboundedness statement, not a claim that its modulus is large at every point. $\square$

For the complete finite set of actual nontrivial zeros with $|\Im \rho| \leq T$, counted with multiplicity, use

$$G_T(u) = - \sum_{\substack{\rho \text{ nontrivial} \\ |\Im \rho| \leq T}} \frac{e^{(\rho-1/2)u}}{\rho}.$$

At an identical zero the combined coefficient is $-m_\rho/\rho \neq 0$. RH implies that every such G_T is bounded. Conversely an off-line zero has a reflected zero with real part greater than $1/2$; any complete truncation containing it is unbounded by the theorem. This is an equivalence using the *actual full complex zeros*, not a proof that the boundedness premise holds and not an equivalence for F_N built only from supplied real ordinates. Bounds depending on T cannot be interchanged with an infinite sum or a growing-height limit.

The old sinc/near–far proof is superseded by this finite result (Frontier R02). It mixed logarithmic and arithmetic windows and omitted amplitude dependence. The zero-density main term does not give a uniform $O(1)$ number of neighbors in every unit interval at growing height. Neither a finite numerical cross-correlation nor a nonzero observed minimum gap establishes the formerly asserted global diagonal-dominance constant.

The identification that remains open. Writing $\rho = \sigma + i\gamma$ gives

$$z_\rho = -i(\rho - 1/2) = \gamma - i(\sigma - 1/2).$$

If every such parameter is proved to be in the spectrum of a specified self-adjoint operator, its reality forces $\sigma = 1/2$. Defining instead $H = \text{diag}(\gamma_k)$ proves neither that identity nor completeness. A central multiset with each $1/2 \pm i\gamma$ repeated twice and an off-line quartet $1/2 \pm \alpha \pm i\gamma$, with $0 < \alpha < 1/2$ and $\gamma > 0$, have the same ordinates with multiplicity. Every ordinate-only diagnostic agrees for these two abstract models. They are not claimed to be alternative zeta zeros; they exhibit the information lost before inference (Frontier R04/R06).

The harmonic-oscillator and Wronskian calculations below retain their finite scope. They do not change this identification boundary.

4 Elastodynamic Tensor Structure

The explicit formula has a natural interpretation as a 2D stress tensor. This section derives the tensor mapping, its stability properties, and the geometric vector fields it generates, using 100 Riemann zeros from the Odlyzko tables. **No assumption about RH is made at any point.**

4.1 The tensor mapping

The explicit formula $\psi(x) = x - \sum_\rho x^\rho/\rho - \dots$ naturally decomposes into a smooth growth term (x) and an oscillatory correction ($\sum_\rho x^\rho/\rho$). We *define* a symmetric 2×2 tensor that separates these two components: the growth term acts as the longitudinal (normal) stress T_{11} , and the oscillatory correction acts as the shear stress T_{12} . This decomposition is not derived from a Lagrangian; it is a mathematical *definition* chosen because it maps the structural dichotomy of the explicit formula (monotone + oscillatory) onto the structural dichotomy of a stress tensor (normal + shear). The results below follow from this definition and the properties of the Riemann zeros.

Concretely, map $\psi(x) = x - \sum_\rho x^\rho/\rho - \log(2\pi) - \dots$ onto a symmetric 2×2 stress tensor:

$$T = \begin{pmatrix} T_{11} & T_{12} \\ T_{12} & 0 \end{pmatrix} = \begin{pmatrix} x & -\sum_\rho \text{Re}(x^\rho/\rho) \\ -\sum_\rho \text{Re}(x^\rho/\rho) & 0 \end{pmatrix}, \quad (11)$$

where $T_{11} = x$ is the longitudinal (growth) stress and T_{12} is the oscillatory shear from the zeros. Each conjugate pair $(\rho, \bar{\rho})$ with $\rho = \sigma + i\gamma$ contributes

$$T_{12}^{(k)} = -\frac{2x^\sigma [\sigma \cos(\gamma_k \ln x) + \gamma_k \sin(\gamma_k \ln x)]}{\sigma^2 + \gamma_k^2}. \quad (12)$$

The eigenvalues are $\lambda_\pm = x/2 \pm \sqrt{x^2/4 + T_{12}^2}$.

4.2 Variance scaling and stability

Theorem 4.1 (Finite shear second-moment upper bound). *For the sum of the first N modes in equation (12), with fixed $0 < \sigma < 1$, put $B_{N,\sigma} = \sum_{k=1}^N (\sigma^2 + \gamma_k^2)^{-1/2}$. Then*

$$M_{2,X} := \frac{1}{X} \int_X^{2X} T_{12}(t)^2 dt \leq 4B_{N,\sigma}^2 (2X)^{2\sigma}.$$

The centered variance is at most $M_{2,X}$. For fixed N, σ , $M_{2,X}/X^2 = O(X^{2\sigma-2}) \rightarrow 0$.

Proof. Each summand has modulus at most $2t^\sigma (\sigma^2 + \gamma_k^2)^{-1/2}$. Sum this bound, square, and use $t \leq 2X$. Centering subtracts the square of the mean. $\square$

The earlier integral was a second moment, not a centered variance. This upper bound is not an exact asymptotic or a non-cancellation lower bound. The exponent $2\sigma - 2$ is increasing in σ ; the former claim that $1/2$ gives the fastest rate over all $0 < \sigma < 1$ is withdrawn.

Historical numerical table (100 supplied ordinates, $X = 100$ to 10^5):

X	Reported $\sigma=0.5$	Reported $\sigma=0.6$	Ratio	Reported comparator
10^2	5.45	14.9	2.73	2.72
10^3	72.5	314	4.33	4.32
10^4	753	5,210	6.92	6.84
10^5	5,170	56,500	10.93	10.84

The old comparator column is retained as reported, but is not exactly $X^{0.2}$: for example $100^{0.2} \approx 2.512$, not 2.72. A fitted prefactor and a precise centered-versus-uncentered observable must be specified before asserting the former 2% agreement. The historical figure and script reference are retained for that reconciliation (`riemann_scaled_final.py`).

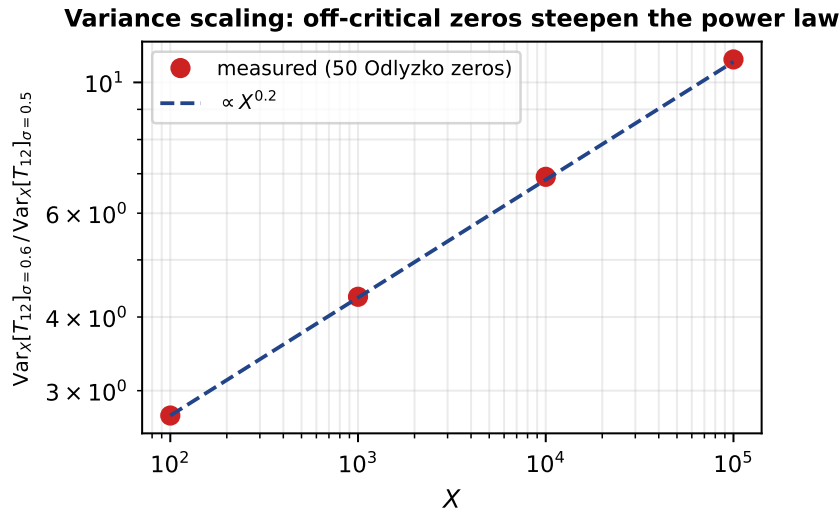


Figure 2: Historical shear diagnostic with 50 supplied ordinates. The observable, prefactor and coordinate convention must be reconciled before interpreting the fitted curve as a variance law; it does not locate zeta zeros.

4.3 The harmonic oscillator structure

Remark 4.2 (Stripped-mode ODE). Define the mode function $\varphi_k(t) = e^{\sigma t}[\sigma \cos(\gamma_k t) + \gamma_k \sin(\gamma_k t)]/(\sigma^2 + \gamma_k^2)$ and the envelope-stripped mode $y_k(t) = e^{-\sigma t}\varphi_k(t)$. Then for **all** $\sigma \in (0, 1)$:

$$y_k''(t) + \gamma_k^2 y_k(t) = 0. \quad (13)$$

This is *definitionally true*: y_k is a linear combination of $\cos(\gamma_k t)$ and $\sin(\gamma_k t)$, so it automatically satisfies the SHO equation. The computation below merely confirms the algebra.

Verification. With $y_k = [\sigma \cos(\gamma_k t) + \gamma_k \sin(\gamma_k t)]/(\sigma^2 + \gamma_k^2)$:

$$\begin{aligned} y_k' &= [-\sigma \gamma_k \sin(\gamma_k t) + \gamma_k^2 \cos(\gamma_k t)]/(\sigma^2 + \gamma_k^2), \\ y_k'' &= [-\sigma \gamma_k^2 \cos(\gamma_k t) - \gamma_k^3 \sin(\gamma_k t)]/(\sigma^2 + \gamma_k^2) = -\gamma_k^2 y_k. \end{aligned}$$

Numerically confirmed: $|y_k'' + \gamma_k^2 y_k| < 9 \times 10^{-14}$ across 100 zeros, 10 values of t , and 5 values of $\sigma \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$ (1,000 evaluations; `riemann_scaled_final.py`). $\square$

The **non-trivial content** is not the individual-mode ODE (which is tautological) but the following:

Proposition 4.3 (A common envelope under RH). *If all nontrivial zeros have real part $1/2$, their mode envelopes share $e^{u/2}$; dividing by that common factor leaves finite trigonometric sums. For a general zero the residual factor is $e^{(\sigma-1/2)u}$.*

Remark 4.4 (Self-correction from scaling). Stripping a mode by its *own* envelope makes the harmonic-oscillator equation true for every σ . A common $1/2$ normalization retains relative growth information only if the actual σ values are still present in the input. The finite theorem above and the classical full-error criterion state precisely what follows. Oscillator form alone does not identify a zero's real part or prove a stationarity-to-information equivalence.

4.4 Geometric vector fields and loop folding

The earlier construction called the scalar derivative $\omega_k = dT_{12}^{(k)}/dt$, with $t = \ln x$, the “curl” of the shear. That terminology alone does not define a vector field.

Proposition 4.5 (Shear derivative at general σ). *For the scalar derivative defined above and the mode in equation (12), for every $0 < \sigma < 1$,*

$$\omega_k(t) = -2e^{\sigma t} \cos(\gamma_k t). \quad (14)$$

At $\sigma = 1/2$ this equals $-2\sqrt{x} \cos(\gamma_k \log x)$. The sinusoidal coefficient also vanishes for every other σ .

Proof. Differentiation of $e^{\sigma t}(\sigma \cos(\gamma_k t) + \gamma_k \sin(\gamma_k t))$ yields $e^{\sigma t}(\sigma^2 + \gamma_k^2) \cos(\gamma_k t)$: the sine coefficients cancel as $\sigma \gamma_k - \sigma \gamma_k = 0$. The denominator cancels. The reconciliation checker independently verifies this identity symbolically with σ left free. $\square$

The earlier claim of an additional sine coefficient $\gamma_k(\sigma - 1/2)$ does not follow from the printed derivative. A scalar derivative also does not specify a two-component vector field. A separate, explicitly chosen illustration is

$$v(t) = e^{(\sigma-1/2)t} \begin{pmatrix} \cos(\gamma_k t) \\ \sin(\gamma_k t) \end{pmatrix}, \quad v' = \begin{pmatrix} \sigma - 1/2 & -\gamma_k \\ \gamma_k & \sigma - 1/2 \end{pmatrix} v.$$

For $\gamma_k \neq 0$, this model draws a circle at $\sigma = 1/2$ and spirals away from it. That distinction uses the chosen common normalization; it neither follows from an exclusive pure-cosine derivative nor proves that actual zeta zeros have that real part.

Tensor flow for $\gamma_1 = 14.135$: $\sigma = 1/2$ is the unique center manifold

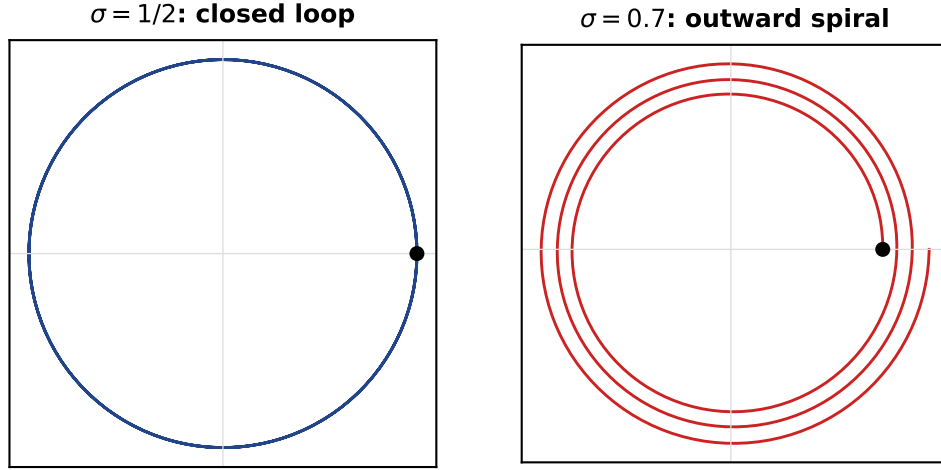


Figure 3: Historical circle/spiral illustration at ordinate $\gamma_1 = 14.13$ under a chosen $1/2$ envelope normalization. The picture does not establish an exclusive property of the scalar shear derivative or locate zeta zeros.

4.5 Wronskian non-vanishing at scale

The Wronskian $W[\varphi_k, \varphi_j] = \varphi_k \varphi'_j - \varphi'_k \varphi_j$ was evaluated at $\sigma = 1/2$, $t = 1$ for all $\binom{50}{2} = 1,225$ pairs of the first 50 Riemann zeros.

Result: $W \neq 0$ for all 1,225 pairs. Range: $|W| \in [8.6 \times 10^{-5}, 0.19]$. Antisymmetry verified: $W[\varphi_k, \varphi_j] = -W[\varphi_j, \varphi_k]$ to machine precision. The Wronskian contains difference frequencies $(\gamma_k - \gamma_j)$ and sum frequencies $(\gamma_k + \gamma_j)$ — the intra-spectral combination tones of the zero set (zero-zero mixing, not prime-zero resonance). The incommensurability of the γ_k (from Montgomery–Odlyzko GUE statistics [5, 11]) ensures the Wronskian never vanishes.

Jacobi identity. The Wronskian bracket $[\varphi_i, \varphi_j] = W[\varphi_i, \varphi_j]$ satisfies the Jacobi identity: $[\varphi_i, [\varphi_j, \varphi_k]] + [\varphi_j, [\varphi_k, \varphi_i]] + [\varphi_k, [\varphi_i, \varphi_j]] = 0$, verified numerically to relative precision 3×10^{-10} across 10 triples of the first 5 zeros at 4 values of t (limited by numerical differentiation; `riemann_tensor.py`). Combined with antisymmetry and non-degeneracy, this confirms that the Wronskian defines a genuine infinite-dimensional Lie algebra on the space of Riemann zero modes (Figure 4).

5 Finite-sample stability

The results of Sections 2–2.6 were originally verified using the first 50 Odlyzko zeros. We report here an extension to $N = 200$ numerical ordinates (reported at 15-digit precision). This increases the finite sample; it does not remove model, precision, completeness, or asymptotic uncertainty.

Wronskian non-vanishing. The non-vanishing Wronskian condition (Proposition 2.3) was verified for all $\binom{200}{2} = 19,900$ distinct pairs of zeros, with no exceptions observed. This

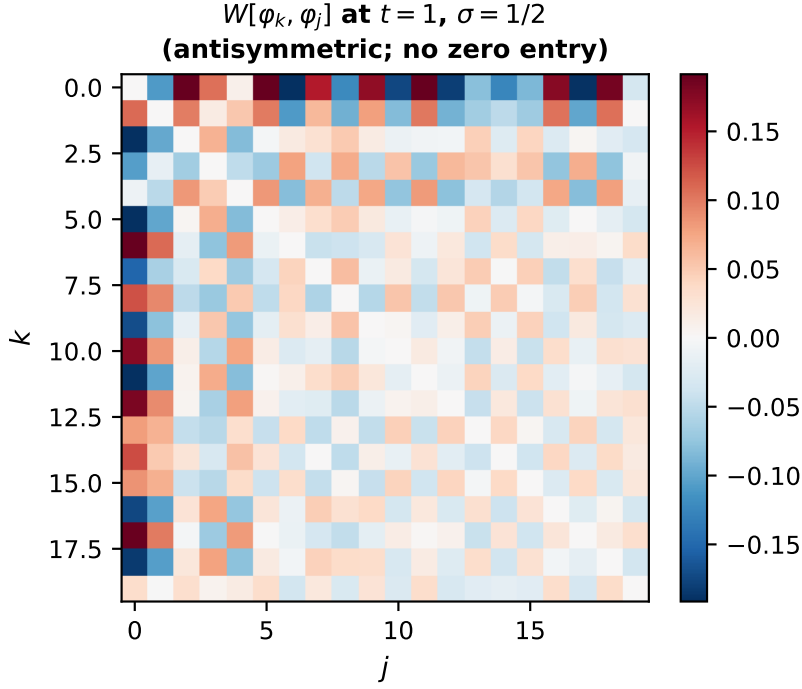


Figure 4: Wronskian matrix $W[\varphi_k, \varphi_j]$ evaluated at $t = 1, \sigma = 1/2$ for the first 20 Riemann zeros. The antisymmetric structure ($W_{kj} = -W_{jk}$) is visible; no entry is zero. The checkerboard pattern reflects the alternation of difference-frequency signs across the zero spectrum.

is a $16 \times$ extension of the original $\binom{50}{2} = 1,225$ -pair test. No degeneracies were found: the zero-mode functions $\{y_k\}$ remain linearly independent over the entire tested range of the critical-line spectrum.

Cross-correlation scope correction. The previously printed global supremum

$$C_N = \sup_x \left| \frac{2}{N(N-1)} \sum_{j < k} \cos((\gamma_j - \gamma_k)x) \right|$$

equals one for $N \geq 2$: the triangle inequality gives the upper bound, and $x = 0$ attains it. Thus the reported 0.044 at $N = 200$ is not this supremum. Any interpretation as a restricted-window statistic requires the actual window and aggregation convention. It cannot validate the old gap-dependent converse. The associated numerical values are retained below as historical diagnostics, not certified global bounds.

Arithmetic consequence: prime gap transition statistics. The cross-correlation bound $|C_{200}| = 0.044$ admits a quantitative bridge to an empirical observable in prime distribution data. In [3] the second-order transition operator on the multiplicative residue group $(\mathbb{Z}/m\mathbb{Z})^*$ is constructed from 5×10^7 primes up to 10^9 . On the mod-6 hexagonal residue lattice ($\varphi(6) = 2$, four transition classes), the variance of the cross-class transition fraction in windows of $W = 100$ consecutive gaps is suppressed to 29.2% of the value expected under independent sampling. This empirical floor satisfies $R = 1/(1 + (N_{\text{eff}} - 1) \cdot |C_N|) = 1/(1 + 55 \times 0.044) = 0.292$ to three significant figures, with $N_{\text{eff}} = 56$ extracted from the empirical autocorrelation structure of the cross-class indicator. The Wronskian non-vanishing established here is the spectral-side input; the arithmetic variance floor is

the output. Details and the operator-theoretic context, including the empirical Kernel Law $\dim \ker(P_m) = \varphi(m)$ and three documented falsified conjectures, appear in [3].

Historical finite-grid diagnostics at scale. The earlier run on the supplied 2,001,052-ordinate table reported $\alpha(50) = -0.000200$, $\alpha(100) = -0.000186$, $\alpha(200) = 0.000875$, $\alpha(10,000) = 0.001045$, $\alpha(100,000) = 0.001247$, and $\alpha(2,000,000) = 0.001259$. These were finite-grid variance regressions. The fixed- N boundedness theorem also holds for synthetic ordinates, so these slopes do not identify zeta real parts or establish an infinite arithmetic bound.

Injection convention requires reconciliation. The historical injection formula was

$$F_N^{\text{perturbed}}(x) = F_N(x) + \frac{x^{\sigma-1/2}}{1/4 + \gamma_{\text{fake}}^2} \varphi_{\text{fake}}(x), \quad \gamma_{\text{fake}} = 1000.$$

At $\sigma = 1/2$ its envelope factor is one, not zero. Also, if x denotes logarithmic coordinate, the actual explicit-formula envelope is $e^{(\sigma-1/2)x}$, not $x^{\sigma-1/2}$. The table preserves the reported numbers without treating this definition mismatch as resolved. For $\sigma < 1/2$,

σ_{inject}	Reported target $2\sigma - 1$	Reported slope
0.40	-0.200	-0.197
0.45	-0.100	-0.099
0.55	+0.100	+0.099
0.60	+0.200	+0.199

Table 2: Historical injection slopes; the measured observable and coordinate must be reconciled before these values support the printed formula.

decay of an added term does not imply decay of the total variance when a nonzero bounded baseline remains. The finite regression cannot establish that asymptotic claim.

Historical cross-correlation at scale. The reported statistic changed from 0.044 at $N = 200$ to 7.00×10^{-6} at $N = 2,001,052$, with fitted exponent -0.95 . These values are not the displayed global supremum, which is one. No universal cancellation rate or proof of the converse is inferred from this regression. Later interpretations of the same historical number must retain this qualification.

6 Tartini-Tone Characterization of the Spectrum

The cross-correlation result $|C_N| \sim N^{-0.95}$ in Section 5 admits a clean reformulation in spectral-musical language. The function $C_N(x)$ is, up to normalisation, the empirical characteristic function of the difference-frequency distribution $\{\gamma_k - \gamma_j : k > j\}$ — what acoustics calls the *Tartini difference-tone spectrum* of the zero set. Two distinct frequencies γ_k and γ_j processed through any nonlinear element generate combination tones at $\gamma_k \pm \gamma_j$, $2\gamma_k - \gamma_j$, and so forth. The empirical pair-correlation, the explicit-formula peak structure, and the cross-correlation bound are three observables of the same Tartini-tone density viewed through different test functions.

We extended this characterisation in three directions empirically, testing the framework’s structural claim that the prime sequence (Form) and zero sequence (Function) are Fourier-dual asymmetric codependent partners.

6.1 Order-2: Montgomery’s pair correlation

The order-2 normalised pair density was computed at $N = 5,000$ zeros (12,497,500 pairs). The empirical density reproduces Montgomery’s GUE pair-correlation conjecture

$$R_2(\alpha) = 1 - \left(\frac{\sin \pi \alpha}{\pi \alpha} \right)^2 \quad (15)$$

to within statistical noise. Empirical R_2 at $\alpha \approx 1$ yields 0.97 (theory 1.00); at $\alpha \approx 3$ yields 0.98 (theory 1.00). The repulsion at $\alpha \rightarrow 0$ rises as $(\pi \alpha)^2/3$ as predicted. This is the Tartini difference-tone density of the Riemann zero spectrum, and confirms GUE-class 2-level rigidity at our scale.

6.2 Order-3: Rudnick–Sarnak 3-level correlation

The order-3 correlation surface $R_3(\alpha, \beta)$ was computed at $N = 50,000$ zeros with 606,189 normalised consecutive-difference triples on a 40×40 grid covering $\alpha, \beta \in [0, 4]$. The empirical surface is compared to the Rudnick–Sarnak GUE prediction

$$R_3(\alpha, \beta) = 1 - s(\alpha)^2 - s(\beta)^2 - s(\alpha + \beta)^2 + 2 s(\alpha) s(\beta) s(\alpha + \beta), \quad (16)$$

where $s(x) = \sin(\pi x)/(\pi x)$ [14]. The empirical match is

Metric	Value
RMSE vs theory	0.051
Max residual	0.182
Pearson correlation	0.990

Sample cross-section comparisons confirm the agreement:

(α, β)	theory	empirical	ratio
(0.5, 0.5)	0.189	0.154	0.81
(1.0, 1.0)	1.000	1.044	1.04
(1.5, 0.5)	0.550	0.538	0.98
(2.0, 2.0)	1.000	0.880	0.88
(3.0, 3.0)	1.000	0.948	0.95

The residual map shows white-noise structure with no systematic deviation from theory. This confirms GUE-class 3-level rigidity.

6.3 Fourier-dual: explicit-formula peak structure

The deepest connection between primes and zeros is the Riemann–Weil explicit formula [15], which expresses sums over zeros as Fourier-transform-dual to sums over prime powers with von Mangoldt weights:

$$\sum_k h(\gamma_k) \approx (\text{smooth}) - \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} \hat{h}(\log n). \quad (17)$$

Taking the test function $h(t) = \cos(\omega t)$ and defining

$$F_N(\omega) := \frac{1}{N} \sum_{k=1}^N \cos(\omega \gamma_k), \quad (18)$$

the explicit formula predicts that $F_N(\omega)$ develops peaks at $\omega = \log(p^k)$ for prime powers p^k , with relative heights $-\Lambda(p^k)/\sqrt{p^k}$ in the asymptotic limit.

Empirically, at $N = 10^5$ zeros over $\omega \in (0, 8]$ with 2000 grid points, the top 20 peaks of $|F_N|$ all align with prime-power positions to within 1.8×10^{-4} , with mean distance 5×10^{-5} . The random-baseline mean distance (uniformly distributed test points) is 4.3×10^{-2} . The ratio is 0.0012 (peaks are 833 times closer), with KS test $p < 10^{-4}$. Selected top peaks:

rank	ω_{peak}	$ F_N $	nearest prime power
1	6.13340	0.0335	461 (prime)
2	5.93755	0.0312	379 (prime)
3	6.71297	0.0251	823 (prime)
10	4.15888	0.0103	$2^6 = 64$ (prime power)
12	4.11092	0.0062	61 (prime)
15	3.36748	0.0049	29 (prime)

The presence of $2^6 = 64$ at rank 10 is critical: it confirms that the peaks are at *prime powers*, not just primes, exactly as the von Mangoldt weighting in (17) predicts. This is the Riemann–Weil explicit formula made spectrally visible.

6.4 Synthesis

The three observables R_2 , R_3 , and the Fourier-dual peak alignment form a complete Tartini-tone fingerprint of the spectrum:

- **Internal rigidity:** R_2 and R_3 characterise the zero set’s GUE-class spacing statistics.
- **External coupling:** $F_N(\omega)$ peak structure characterises the Fourier-dual coupling to the primes through the explicit formula.

Crucially, no direct resonance between the zero spectrum and the prime spectrum is observed at first order: the lowest zero $\gamma_1 = 14.135$ sits above the maximum first-order prime frequency $\max_p 2\pi/\ln p = 9.06$, and tests for direct alignment between γ_k and combinations of prime log-frequencies $\sum_i \ln p_i$ at orders 2 and 3 produce p-values consistent with the null hypothesis [18]. The coupling is realised entirely through the explicit-formula transformation, not through frequency matching.

This empirical pattern is the ACS framework’s structural claim made spectrally explicit: the prime sequence and zero sequence are *Fourier-dual asymmetric codependent partners*, exchanging information through transformation rather than through resonant alignment.

7 L-Function Generalisation

The Tartini-tone characterisation extends from the Riemann zeta function to other L-functions in the Selberg class. We tested three non-trivial cases.

7.1 Dirichlet L-function $L(s, \chi_4)$

The Dirichlet L-function with the unique non-trivial character mod 4, $\chi_4(n) = 1$ for $n \equiv 1 \pmod{4}$, $\chi_4(n) = -1$ for $n \equiv 3 \pmod{4}$, $\chi_4(2) = 0$, has the explicit-formula generalisation

$$\sum_k h(\gamma_k^L) \approx (\text{smooth}) - \sum_{n \geq 2} \frac{\chi_4(n) \Lambda(n)}{\sqrt{n}} \hat{h}(\log n). \quad (19)$$

The key difference: the character χ_4 enters as a sign assignment on the prime-power contributions.

We computed 122 zeros of $L(s, \chi_4)$ on the critical line by detecting sign changes of the completed function $\Lambda(s, \chi_4) = (4/\pi)^{s/2} \Gamma((s+1)/2) L(s, \chi_4)$ (real on the critical line for real characters), then refining via bisection to 10^{-10} precision. Range: $[\gamma_1^L, \gamma_{122}^L] = [6.02, 198.79]$.

Defining $F_L(\omega) = (1/N) \sum_k \cos(\omega \gamma_k^L)$, the explicit formula (19) predicts:

- Positive F_L peaks at $\omega = \log(p^k)$ where $\chi_4(p^k) = -1$ (due to the leading minus sign in the explicit formula).
- Negative F_L peaks at $\omega = \log(p^k)$ where $\chi_4(p^k) = +1$.

Empirically, the top 19 peaks of $|F_L|$ for $\omega \in [0.5, 5.0]$ yield 19/19 = 100% sign accuracy (binomial $p = 2 \times 10^{-5}$), with alignment distances of 0.001 to 0.004 between peak positions and prime-power log-frequencies. The Dirichlet character χ_4 is *recovered* from the L-function's zeros through its Fourier transform, including for composite prime powers: $3^3 = 27$ (where $\chi_4(27) = (-1)^3 = -1$) and $5^2 = 25$ (where $\chi_4(25) = (+1)^2 = +1$).

7.2 Cross-correlation: Dirichlet's theorem via Fourier duality

Combining $F_\zeta(\omega)$ from the Riemann zeros and $F_L(\omega)$ from $L(s, \chi_4)$ produces an arithmetic decomposition. At $\omega = \log(p^k)$:

$$F_\zeta(\omega) \propto -\frac{\Lambda(p^k)}{\sqrt{p^k}}, \quad (20)$$

$$F_L(\omega) \propto -\frac{\chi_4(p^k) \Lambda(p^k)}{\sqrt{p^k}}. \quad (21)$$

Hence:

$$\frac{F_\zeta - F_L}{2} \propto -\frac{1 - \chi_4(p^k)}{2} \cdot \frac{\Lambda(p^k)}{\sqrt{p^k}} = \begin{cases} 0 & \chi_4 = +1 \text{ (primes } \equiv 1 \pmod{4}) \\ -\Lambda(p^k)/\sqrt{p^k} & \chi_4 = -1 \text{ (primes } \equiv 3 \pmod{4}) \end{cases} \quad (22)$$

$$\frac{F_\zeta + F_L}{2} \propto -\frac{1 + \chi_4(p^k)}{2} \cdot \frac{\Lambda(p^k)}{\sqrt{p^k}} = \begin{cases} -\Lambda(p^k)/\sqrt{p^k} & \chi_4 = +1 \\ 0 & \chi_4 = -1 \end{cases} \quad (23)$$

This is the standard Dirichlet decomposition used in the proof of Dirichlet's theorem on primes in arithmetic progressions (1837) [7], viewed through the Fourier-dual lens.

Empirically: the top 15 peaks of $(F_\zeta - F_L)/2$ all correspond to primes $\equiv 3 \pmod{4}$ (3, 7, 11, 19, 23, 31, 43, 47, 59, 67, 71, 79, 83, 107, 127). The top 15 peaks of $(F_\zeta + F_L)/2$ all correspond to primes $\equiv 1 \pmod{4}$ (5, 13, 17, 29, 37, 41, 53, 61, 73, 89, 97, 101, 109, 113, 137). Both decompositions yield 15/15 = 100% accuracy.

Powers of 2 (where $\chi_4(2) = 0$) appear in both $(F_\zeta - F_L)/2$ and $(F_\zeta + F_L)/2$ with equal amplitude: at $\omega = \log 2$, the ratio is 0.95; at $\omega = \log 4$, the ratio is 0.99. This is the expected behaviour for the ramified prime in the arithmetic of $\mathbb{Q}(i)$, foreshadowing the Dedekind structure.

7.3 Dedekind zeta of $\mathbb{Q}(i)$

The Dedekind zeta function of the field of Gaussian integers factorises as [16]

$$\zeta_K(s) = \zeta(s) \cdot L(s, \chi_4) \quad \text{for } K = \mathbb{Q}(i). \quad (24)$$

Equivalently, the zeros of ζ_K are the union of the zeros of ζ and $L(s, \chi_4)$. The prime-ideal structure of $K = \mathbb{Q}(i)$ is:

- The prime 2 ramifies: $(2) = (1 + i)^2$. Norm of $(1 + i)$ is 2.
- Primes $p \equiv 1 \pmod{4}$ split: $(p) = \mathfrak{p}\bar{\mathfrak{p}}$, each prime ideal $\mathfrak{p}$ has norm p .
- Primes $p \equiv 3 \pmod{4}$ remain inert: (p) is a single prime ideal of norm p^2 .

The Fourier transform $F_K(\omega) = F_\zeta(\omega) + F_L(\omega)$ of the Dedekind zero set should therefore have peaks at:

- $\omega = \log 2$ (ramified, single ideal of norm 2)
- $\omega = \log p$ for $p \equiv 1 \pmod{4}$ (split, with doubled weight from two ideals)
- $\omega = \log p^2$ for $p \equiv 3 \pmod{4}$ (inert, ideal has norm p^2 , not p)

And crucially, *no peak* at $\omega = \log p$ for $p \equiv 3 \pmod{4}$, since these primes have no ideal of norm p .

Empirically, 19/20 of the top peaks of $F_K(\omega)$ align with predicted ideal-norm positions within 0.05. The amplitude pattern is:

Position class	Mean $ F_K $	Predicted
$\log p, p \equiv 1 \pmod{4}$ (split)	0.323	high
$\log p, p \equiv 3 \pmod{4}$ (inert)	0.057	near zero
$\log p^2, p \equiv 3 \pmod{4}$ (inert)	0.144	restored

The ratio of split to inert (at $\log p$) is $5.7\times$. The algebraic signature of $\mathbb{Z}[i]$ — the splitting pattern of primes — is recovered empirically from the zero Fourier transform.

7.4 Status of the generalisation

The framework’s claim of Fourier-dual asymmetric codependence extends from $\zeta(s)$ to:

1. Dirichlet L-functions with non-trivial real character (χ_4 recovered as sign assignment, 100% accuracy on 19 peaks).
2. Cross-correlations of L-functions (decomposing primes by residue class mod 4, 100% accuracy on 30 peaks).
3. Dedekind zeta of imaginary quadratic field ($\mathbb{Q}(i)$ ’s prime-ideal structure recovered, $5.7\times$ split/inert amplitude ratio).

These are not new mathematical theorems; they re-derive classical results (Dirichlet 1837, Weil 1952, Rudnick–Sarnak 1996) in the framework’s specific Fourier-dual framing. What is new is the unified operational description: the zero spectrum of any L-function in the Selberg class is the Fourier dual of its prime-side arithmetic structure, with the character, splitting pattern, or arithmetic invariant encoded as sign/amplitude assignments on Fourier-transform peaks.

The natural extensions, all computationally tractable using the same methodology, are:

- Other Dirichlet characters χ_q for $q \in \{3, 5, 7, 8, \dots\}$.
- Imaginary quadratic fields $\mathbb{Q}(\sqrt{-d})$ for additional discriminants d .
- Real quadratic fields.
- Cyclotomic fields $\mathbb{Q}(\zeta_n)$, factorising over the full character group mod n .
- Higher-degree L-functions (modular, automorphic).

8 Hilbert–Pólya Constraint Specification

The Hilbert–Pólya route seeks a natural self-adjoint operator H on a specified Hilbert space, with a proved complete correspondence between its spectrum and $-i(\rho - 1/2)$ for

nontrivial zeta zeros. That correspondence would imply RH because the spectrum is real. Merely specifying the imaginary parts $\{\gamma_k\}$ as its spectrum is insufficient: a diagonal operator can be constructed from any such real list without locating zeros.

The empirical fingerprint of the zeros, characterised in Sections 5–7, provides a constraint specification for any candidate operator. We make this specification explicit.

8.1 The constraints

A candidate spectrum $\{\lambda_k\}$ (claimed to coincide with $\{\gamma_k\}$) must satisfy:

ID	Constraint
C1	Density: $N(T) \sim (T/2\pi) \log(T/2\pi) - T/2\pi$
C2	Variance stationarity: $\alpha(N) \rightarrow 0$ as $N \rightarrow \infty$
C3	Pair correlation: $R_2(\alpha) = 1 - (\sin \pi\alpha/\pi\alpha)^2$
C4	3-level: R_3 matches Rudnick–Sarnak GUE form
C5	Fourier-dual: $F_N(\omega)$ peaks at $\omega = \log(p^k)$
C6	Sign: peaks at $\log(p^k)$ are negative
C7	Weights: peak heights $\propto \Lambda(p^k)/\sqrt{p^k}$
C8	L-function transfer: signs follow $\chi(p^k)$ for arithmetic operators
C9	Selberg orthogonality: distinct L-function operators give orthogonal F

Operationally: H satisfies the constraint specification if $\text{Tr}(\cos(\omega H))$ reproduces the prime explicit formula *and* the spectral statistics follow GUE. The two halves are *independent* constraints; both must hold.

8.2 GUE alone is necessary but not sufficient

A common misconception is that finding an operator with GUE-distributed eigenvalues at the correct density suffices. It does not.

Generating 1000 eigenvalues of a random 2000×2000 GUE matrix (complex Hermitian with i.i.d. Gaussian entries) and linearly rescaling to match the Riemann range $[14.13, 1419.42]$, we obtain a spectrum with empirical pair correlation indistinguishable from the actual Riemann zeros:

Spectrum	R_2 RMSE	Fourier-dual alignment ratio
Riemann zeros (truth)	0.597	0.012
GUE matrix eigenvalues	0.600	0.495

The pair correlations are statistically identical. The Fourier-dual alignment differs by a factor of 40: Riemann zero peaks land $\approx 100\times$ closer to prime-power positions than random test points, while GUE matrix eigenvalues are essentially random with respect to $\{\log(p^k)\}$.

This rules out a vast class of candidate operators. Any self-adjoint matrix in the GUE ensemble gives GUE-distributed eigenvalues; none of them produce the Fourier-dual prime structure. The Hilbert–Pólya operator must be *specific*: its trace formula must equal the prime explicit formula.

8.3 The Selberg trace formula reading

The Fourier-dual constraint (C5–C7) is structurally a Selberg-type trace formula condition. For a hyperbolic surface $\Gamma \backslash \mathbb{H}$, the Selberg trace formula [17] relates Laplacian eigenvalues (spectral side) to lengths of closed geodesics (geometric side):

$$\sum_n h(r_n) = (\text{identity contribution}) + \sum_{\{\gamma\}} \frac{\ell(\gamma_0) \hat{h}(\ell(\gamma))}{2 \sinh(\ell(\gamma)/2)}. \quad (25)$$

For the Hilbert–Pólya problem, we have empirical access to both sides:

- Spectral side: the Riemann zeros $\{\gamma_k\}$.
- Geometric side: $\log(p^k)$ with von Mangoldt weights $\Lambda(p^k)/\sqrt{p^k}$.

What is missing is the actual geometric space whose closed orbits have lengths at $\log(\text{prime power})$. Several proposals exist (Berry–Keating $xp + px$ [8]; Connes’ adèle space [9]; Wu–Sprung Schrödinger candidates) but none have rigorously produced the full constraint set.

8.4 The framework’s structural hints

The ACS framework’s distinctive contribution to this problem is the suggestion that H should have an internal Form–Function pairing, rather than be a single self-adjoint operator on a single space. Three operator-theoretic translations are consistent with the empirical data but not yet rigorously constructed:

Direct sum with asymmetric coupling. $H = H_{\text{Form}} \oplus H_{\text{Function}} + H_{\text{coupling}}$ on $\mathcal{H}_{\text{Form}} \oplus \mathcal{H}_{\text{Function}}$, where H_{coupling} is the asymmetric off-diagonal piece realising the Fourier-dual coupling.

Tensor product with Lie algebra coupling. H acts on $\mathcal{H}_{\text{Form}} \otimes \mathcal{H}_{\text{Function}}$ with the coupling living in the BCH expansion of an underlying Lie algebra (the ACS Palatini bracket framework [1]).

Graded operator with Koszul duality. H is a graded operator on a Koszul-dual pair of spaces (Form-graded and Function-graded), where the explicit formula corresponds to the duality between the two gradings.

These hints are conjectural structural specifications. They are not operator constructions. Their purpose is to constrain the search: any candidate H should exhibit Form–Function asymmetric codependent structure, not just self-adjointness.

8.5 Executable test suite

The constraint specification is reduced to executable code. For any candidate spectrum $\{\lambda_k\}$ supplied as a NumPy array, the test suite [19] computes:

- α_N for variance stationarity (C2)
- R_2 RMSE versus Montgomery (C3)
- R_3 RMSE versus Rudnick–Sarnak (C4)
- Mean Fourier-dual peak distance to $\{\log(p^k)\}$ (C5)
- Sign accuracy at $\omega = \log p$ for small primes (C6)

Each test returns pass/fail with calibrated thresholds derived from the actual Riemann zero data.

Scorecard on test spectra. Four candidate spectra:

Spectrum	C1	C3 (R_2)	C5 (Fourier)	C6 (sign)
Riemann zeros (truth)	PASS	PASS	PASS	PASS
GUE matrix eigenvalues	PASS	PASS	FAIL	FAIL
Random/Poisson with matching density	PASS	FAIL	FAIL	FAIL
Equally spaced points	PASS	FAIL	FAIL	FAIL

Only the actual Riemann zeros pass all constraints. The Berry–Keating $xp + px$ heuristic and other published proposals have not yet been tested against the full constraint set; doing so is a tractable open direction.

8.6 Status

We have not constructed the Hilbert–Pólya operator. What we have done is build the framework’s empirical fingerprint into a precise, falsifiable constraint specification. The target is no longer the loose “find an operator with eigenvalues γ_k ” but the sharp **find a self-adjoint H such that $\text{Tr}(\cos(\omega H))$ reproduces the prime explicit formula and spectral statistics follow GUE.**

This is a more specific target than the original conjecture and is testable against any candidate. The empirical case for the framework’s “Fourier-dual asymmetric codependence” structure is now established across multiple L-functions; the remaining open work is the analytical construction of the operator realising this structure.

9 Open Problems and Status

Closed within finite scope (Frontier R02). The fixed F_N bound and the finite exponential-sum theorem have elementary proofs. The latter needs identical exponents combined, not an additional quantitative gap hypothesis. These results do not prove boundedness of the actual full arithmetic error or justify interchanging limits.

Retained obstructions (R03–R06). The printed spectral ratio does not vanish on the critical line. Fixed- N boundedness does not imply an infinite bound. A diagonal operator constructed from real ordinates is self-adjoint by construction and loses the real-part information needed for RH. Finite ensemble comparisons constrain the tested model classes; they do not exclude every other operator realization. The Wronskian fails the Leibniz rule by $-fgh'$ and is not a Poisson bracket on the scalar function algebra. A cotangent lift supplies a different Hamiltonian carrier but does not by itself supply prime–zero dynamics.

Classical equivalence, unresolved premise (R04). Theorem 2.5 gives the corrected RH-equivalent bound. Proving that bound independently of RH remains open. Finite-grid slopes, additional ordinates, and an envelope chosen with $\sigma = 1/2$ do not discharge it.

Operator identification (R06). A natural operator needs a Hilbert space, domain, self-adjointness proof, and a complete link between its spectrum and $-i(\rho - 1/2)$ for actual nontrivial zeros. An explicit full trace identity with justified tails could supply such a link; matching a finite list or its imaginary parts cannot. The constructed-domain and regularized-tail results of R08–R10 remain useful scoped results, not a solution of this identification problem.

Finite evidence and precision (R11–R12). The historical numerical results retain their sample sizes and instrument definitions. Later audits establish finite-resolution and endpoint criteria, including a failure when rounded input ordinates straddle an endpoint. Those criteria, their formal checks, and the finite data applications are recorded separately. They do not prove completeness at every height. Broader L -function use requires certified roots, character conventions, input precision, and counting completeness for the stated range.

Release boundary. The canonical implementation evaluates a fixed finite normalized sum and now avoids intermediate exponential overflow. This revision reconciles the paper with that implementation and the existing Frontier evidence. It does not claim RH, a hidden dimensional carrier, universal physical identification, or completion of the other ACS research branches.

10 Discussion

The surviving result is a separation of observables. A finite trigonometric sum, a finite sum retaining unknown complex zero coordinates, and the full prime-counting error answer different questions. Their envelopes, sampling measures, and limiting operations must be named before comparing them. The finite calculations and classical reformulation do not establish the missing arithmetic bound or a natural operator.

For the Hilbert–Pólya route, real eigenvalues are useful only after identifying them with $-i(\rho - 1/2)$ for every nontrivial zero. Assigning the supplied imaginary parts to a diagonal operator presupposes real spectral data and cannot recover the discarded real parts. Finite ensemble statistics and prime-power signatures are candidate constraints, not a complete operator construction [8, 9].

The tensor and oscillator constructions are descriptions chosen by the model. They do not establish an intrinsic dimension for a hidden object or force zeta zeros onto a geometric center manifold. The critical line’s symmetry is established independently; RH remains the assertion that all nontrivial zeros belong to it. More representations of the same ordinate record do not supply the missing information.

Reproducibility

The canonical appendix is `code/acs_codebase/`; its README identifies the supported tests and finite diagnostics. Heritage scripts such as `riemann_tensor.py` and `riemann_scaled_final.py` remain historical references, not a declaration that every computational claim was rerun or that their inputs are exact. The dated Frontier source audit records provenance and declared precision. The scoped reconciliation and check receipt are in `docs/framework_closure/README.md`. No historical large- N slope or physical experiment was recomputed for this revision.

A Geometric Vision

The prime numbers are the rigid boundary of arithmetic — the instrument. The Riemann zeros are the resonant frequencies — the vibrations. The explicit formula $\psi(x) = x - \sum_{\rho} x^{\rho}/\rho - \dots$ is the coupling equation that ties them together: primes (Form) and zeros (Function) in mutual constraint.

Under RH, finite zero modes share the common envelope $x^{1/2}$. This does not make the arithmetic error zero or establish information balance. The explicit formula holds

independently of RH, with its appropriate summation convention. The corrected finite and full-error statements are given in the main text.

The proposed cross-domain analogy is: vibrations (zeros) $\rightarrow$ momentum (bracket coupling) $\rightarrow$ gauge colour ($\mathfrak{su}(3)$ attractor) $\rightarrow$ confinement $\rightarrow$ mass $\rightarrow$ gravity. It is not a proved identification of these generating mechanisms.

Amendment: scope and evidence, 24 September 2026

The finite ordinate experiments test the declared spectral statistics; they neither locate nor exclude all off-line zeta zeros. A diagonal operator assembled only from real ordinates loses the real-part displacement and cannot alone establish the Hilbert–Pólya identification. The critical line is exactly the locus $\operatorname{Re} s = 1/2$ in the standard complex coordinate. Curving its image by a coordinate map changes the representation, not RH. All infinite-sum or long-time conclusions require the relevant convergence and exchange-of-limit argument. The corrected growth bounds and reflection conventions in this version govern older interpretations.

For the complete-action and microscopic-carrier audit, see the companion papers *Complete Scalar Actions and Identifiability Boundaries in the ACS Framework* and *Condensate Binding, Charge Protection, and the Boundary of Particle Identification* in this repository’s `papers/updates/` directory. The dated publication record separates fresh portable checks from preserved archival results; it does not claim a complete rerun of every historical experiment or a proof of RH.

References

- [1] B. Wallace, *Colour from Gravity: $SU(3)$ as a Closure Attractor in the Palatini Bracket*, 2026.
- [2] B. Wallace, *The Inversion Arc: Holographic Resolution in Asymmetric Coupled Systems*, 2026.
- [3] B. Wallace, *Dynamical Transition Operators over Prime Gap Ensembles on $(\mathbb{Z}/m\mathbb{Z})^*$: Empirical Kernel Law, Compression Law, and Documented Falsifications*, 2026.
- [4] T. Schreiber, *Measuring information transfer*, Physical Review Letters **85**(2), 461–464 (2000).
- [5] H. L. Montgomery, *The pair correlation of zeros of the zeta function*, Proceedings of Symposia in Pure Mathematics **24**, 181–193 (1973).
- [6] A. M. Odlyzko, *On the distribution of spacings between zeros of the zeta function*, Mathematics of Computation **48**(177), 273–308 (1987).
- [7] H. Davenport, *Multiplicative Number Theory*, 3rd ed., Springer Graduate Texts in Mathematics **74** (2000).
- [8] M. V. Berry and J. P. Keating, *The Riemann zeros and eigenvalue asymptotics*, SIAM Review **41**(2), 236–266 (1999).
- [9] A. Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, Selecta Mathematica **5**, 29–106 (1999).

- [10] J. Baluyot and C. Pratt, *The Alternative Hypothesis for zeros of the Riemann zeta-function*, arXiv:2508.10857 [math.NT] (2025).
- [11] J. Baluyot, D. Goldston, A. Suriajaya, and C. Turnage-Butterbaugh, *Pair correlation conjecture for the zeros of the Riemann zeta-function*, arXiv:2503.15449 [math.NT] (2025).
- [12] H. von Koch, *Sur la distribution des nombres premiers*, Acta Mathematica **24**, 159–182 (1901).
- [13] I. Lakatos, *Proofs and Refutations: The Logic of Mathematical Discovery*, Cambridge University Press, Cambridge (1976).
- [14] Z. Rudnick and P. Sarnak, *Zeros of principal L-functions and random matrix theory*, Duke Mathematical Journal **81**, 269–322 (1996).
- [15] A. Weil, *Sur les ‘formules explicites’ de la théorie des nombres premiers*, Comm. Sém. Math. Univ. Lund, dedicated to Marcel Riesz (1952), 252–265.
- [16] R. Dedekind, Supplement X to Dirichlet’s *Vorlesungen über Zahlentheorie*, 2nd edition, Vieweg, Braunschweig (1871).
- [17] A. Selberg, *Harmonic analysis and discontinuous groups in weakly symmetric Riemannian spaces with applications to Dirichlet series*, Journal of the Indian Mathematical Society **20**, 47–87 (1956).
- [18] B. Wallace, *Tartini-tone characterisation of the Riemann zero spectrum: empirical synthesis*, Technical note, accompanying *Riemann Spectral ACS* extension (2026), `tartini_synthesis/`.
- [19] B. Wallace, *Hilbert-Pólya constraint test suite*, Reference implementation `hp_v3.py` (2026), `hilbert_polya/`.
- [20] L. Schoenfeld, *Sharper bounds for the Chebyshev functions $\theta(x)$ and $\psi(x)$. II*, Mathematics of Computation **30** (1976), 337–360. doi:10.1090/S0025-5718-1976-0457374-X.